

Submission Date: 8th April 2026

Instructions:

- Show all mathematical steps for derivations and weight updates.
- Clearly label all diagrams (Perceptron models, decision boundaries, etc.).
- Cite your sources (Haykin or Fausett) for any definitions or historical context.
- This assignment must be typed, formatted in APA style, and include a bibliography.
- **Mathematical Rigor:** Ensure you understand the vector-matrix multiplication involved in processing inputs.
- **Diagrams:** Use standard symbols for neurons (circles) and connections (lines/vectors).
- **Historical Context:** Reference Simon Haykin's *Neural Networks and Learning Machines* to explain the transition from early M-P models to modern learning architectures.
- **All work should be handwritten**

SECTION 1: Foundations and Biological Inspiration (15 Marks)

1. Biological vs. Artificial (8 Marks):

- Compare the biological neuron (dendrites, soma, axon, synapses) with the artificial neuron model (inputs, summation, activation function, weights).
- Explain why the "McCulloch-Pitts" model is considered the foundational stone of neural computing. What were its primary limitations?

2. Pattern Recognition Fundamentals (7 Marks):

- Define **Feature Vector** and **Feature Space**.
- Explain the role of **Feature Extraction** in dimensionality reduction. Why is it computationally expensive to classify high-dimensional raw data directly without first extracting features?

SECTION 2: Perceptron Theory and Linear Separability (15 Marks)

1. Architecture (8 Marks):

- Provide a detailed diagram of a Single-Layer Perceptron.
- Derive the mathematical output equation:
$$y = \phi(\sum_{i=1}^n w_i x_i + b)$$

. Define each variable in the equation.

2. Linear Separability (7 Marks):

- Discuss the geometric interpretation of a perceptron as a **Hyperplane** in a feature space.
- Contrast **Linearly Separable** vs. **Non-Linearly Separable** problems. Use the XOR problem as a proof of why a single-layer perceptron fails in non-linear classification.

SECTION 3: Learning Algorithms and Adaptation (18 Marks)

1. Weight Adjustment (10 Marks):

- Explain the **Perceptron Learning Algorithm**. If the target output is d and the actual output is y , write the weight update formula:
$$\Delta w = \eta(d - y)x.$$
- Explain the role of the **Learning Rate (η)** in this process. What happens if η is too large or too small?

2. Hebbian vs. Delta Rule (8 Marks):

- Explain the **Hebbian Learning rule** ("Neurons that fire together, wire together").
- Contrast this with the **Widrow-Hoff (Delta) rule**. Why is the Delta rule considered more effective for minimizing error compared to the basic Perceptron rule?

SECTION 4: Classification and Statistical Methods (12 Marks)

1. Statistical Approaches (6 Marks):

- Describe how **Bayesian Statistical Classification** handles uncertainty in pattern recognition.
- Briefly compare **Deterministic Techniques** (e.g., threshold-based) vs. **Statistical Techniques** in terms of how they handle noise in input data.

2. Neural Processing (6 Marks):

- Elaborate on "Elementary Nets" and how they aggregate to form complex architectures.
- Define **Adaptation** in the context of neural networks. Why is the ability to adapt to new, unseen data considered the "hallmark" of a functional neural network?

Distribution of Marks Summary

Section	Topic	Marks
Section 1	Foundations & Bio-Inspiration	15 Marks
Section 2	Perceptron & Separability	15 Marks

**Section
3**

Learning Algorithms

18 Marks

**Section
4**

Statistical Methods & Adaptati
on

12 Marks

TOTAL

60 MARKS